Reference-Free Reliability Scores Cannot See Corrupted Evidence: A Controlled Test for Numerical Retrieval-Augmented QA in Finance

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Abstract

Reference-free reliability scores for retrieval-augmented generation judge a response against the evidence that was retrieved, without the gold answer. Their structural limit is plain: if the evidence is wrong, a faithful answer is still wrong. We test how far that limit bites, and what gets past it, in numerical financial question answering (FinQA and TAT-QA; Qwen2.5 3B, 7B and 14B and Llama-3.1 8B, up to 200 items per run). When a controlled share of the numbers in the evidence is replaced, accuracy falls to near zero, while numeric-overlap support, entailment support, lexical groundedness, a model-based faithfulness judge, a hallucination flag and verbalized confidence stay close to chance at telling corrupted from clean evidence (AUROC at most 0.60 in any run; pooled numeric-overlap support 0.54 on FinQA and 0.52 on TAT-QA, the judge 0.50 and 0.50). Realistic errors that only misplace real numbers (a swapped period or row) cost up to 54 points of accuracy on TAT-QA and are invisible to every response-side score, and also to support measured against a clean copy of the *numbers* (AUROC 0.43 to 0.52), because every number is still genuine. The results hold for two generator families (Qwen2.5-7B and Llama-3.1-8B). What does see them is a cell-level check against a trusted structured copy (AUROC 0.87 to 0.99); checks that need only the evidence work weakly (redundancy, 0.66 to 0.68 on scaled numbers, 0.50 on swaps) or barely at all (arithmetic self-consistency, AUROC 0.42 to 0.55). The same response-side measures are also weak and model-dependent as detectors of wrong answers and of real retrieval misses. Along the way we correct a scoring flaw and an invalid noise manipulation in an earlier version of this work. Code, per-response outputs and a pipeline that reproduces every result on free CPU runners are public.

1 Introduction

Retrieval-augmented generation (RAG) is the standard way to ground a language model's answers about a document, and in finance the answer is usually a number: a ratio, a change, a total. A wrong number that is stated with confidence is costly, so a practical question is whether the system can tell when its own answer is not to be trusted, without the gold answer. A family of *reference-free* scores tries to do this. They check whether the numbers in the response appear in the retrieved evidence, whether the evidence entails the response, or whether the model's stated confidence matches its accuracy (Es et al., 2024; Saad-Falcon et al., 2024; Min et al., 2023). Several such signals are often folded into one composite, so that a single figure summarizes reliability.

This paper examines one such composite for numerical RAG, the Overall Numerical Reliability Score (ONRS), which weights accuracy, numeric evidence support, a hallucination rate and the calibration of verbalized confidence. We ask a narrow question: *when the retrieved evidence is wrong, does a reference-free measure notice?* The question has a structural answer that we state first and then test. A measure that compares the response with the evidence *that was supplied* can only report whether the response is faithful to that evidence. If the evidence itself has been corrupted, a faithful response is still wrong, and the measure has no way to see it.

The test needs a manipulation that actually degrades the model. Our first attempt did not have one. We began by adding unrelated sentences to the evidence and tracking the composite against the amount added, and we initially read the flat result as a failure of the composite. A manipulation check at larger scale showed that the added sentences did not hurt the model at all, so that experiment could not test anything (Section 5). We therefore replace

a controlled share of the numbers in the retrieved evidence with wrong ones. This does hurt: accuracy falls to near zero at full corruption. It also lets us ask which measures move.

Why this is not as obvious as it sounds. That a score defined relative to the supplied evidence cannot see wrong evidence follows from its definition. What does not follow is how the scores behave in practice, whether anything cheap gets past the limit, and which errors the obvious fix handles. We find that the obvious fix, comparing the numbers of the evidence with a trusted copy, handles scaled numbers perfectly and the realistic misplaced-number errors not at all, and that what handles both needs a trusted *structured* copy.

Contributions.

- A controlled corrupted-evidence test of reference-free reliability measures, with a model-based faithfulness judge among them, on FinQA and TAT-QA and four generators from two families (Qwen2.5 3B, 7B and 14B, Llama-3.1 8B; 100 to 200 items per run). Measures that use only the response and the supplied evidence barely distinguish a fully corrupted case from a clean one (AUROC at most 0.60 in any run). A score that compares the response with the *uncorrupted* evidence can (AUROC 0.84 to 0.94), but it needs the reference we are trying to avoid.
- Two realistic error types that misplace real numbers (a swapped period, a swapped row). They cost accuracy (up to 54 points on TAT-QA), are invisible to every response-side measure, and are invisible to the number-level clean reference that caught scaled numbers (AUROC 0.43 to 0.52), for both generator families.
- A comparison of checks on the evidence itself: redundancy (weak, only for scaled numbers), arithmetic self-consistency (AUROC 0.42 to 0.55, at or near chance), numbers against a trusted copy (perfect for scaled numbers, chance for swaps) and table cells against a trusted structured copy (AUROC 0.87 to 0.99 for the swaps).
- As detectors of wrong answers and of real retrieval misses the response-side measures are weak and model-dependent: near chance when pooled on FinQA, modestly informative on TAT-QA, with effects that change sign between smaller and larger models.
- A correction to a scoring flaw in this line of work (the model's own confidence statement counted as unsupported evidence), a demonstration that an evidence-noise manipulation we first used did not degrade the model, and that five-point correlations and weight sweeps are too unstable to rank composites.
- All code, per-response outputs and the free CPU pipeline that produced them are public.[1]

2 Related Work

Reference-free evaluation of RAG. RAGAs and ARES score faithfulness, answer relevance and context relevance, using a language model as judge or a fine-tuned classifier (Es et al., 2024; Saad-Falcon et al., 2024). Their faithfulness component checks the answer against the context that was retrieved. That is the property whose limit we examine: it is silent about whether the context is right. RAGAs-style faithfulness splits the answer into claims and has a language model judge each against the context (Es et al., 2024; Zheng et al., 2023); we include such a judge as a baseline (Section 3.3). Atomic-claim scoring (Min et al., 2023) and sampling-consistency checks (Manakul et al., 2023) target hallucination in open-ended text. Natural-language-inference models have been used to score factual consistency between a source and a generated text (Maynez et al., 2020; Laban et al., 2022; Zha et al., 2023); we use an NLI cross-encoder as one of our measures.

Confidence and calibration. Expected calibration error (Guo et al., 2017) summarizes the gap between confidence and accuracy. Language models can verbalize a confidence (Kadavath et al., 2022; Xiong et al., 2024), but the figure is often poorly informative; we find the same for the small models studied here.

Financial numerical QA. FinQA (Chen et al., 2021) asks multi-step arithmetic questions over earnings-report pages, with annotated gold evidence. TAT-QA (Zhu et al., 2021) pairs tables with paragraphs and includes arithmetic, span and counting questions. Both are usually evaluated with gold evidence supplied; we use them in a retrieval setting (Lewis et al., 2020) where retrieval errors occur naturally.

[1] https://github.com/aaronsunny007/Financial-NRF

3 Setup

3.1 System

Each report (paragraphs and table rows) is split into chunks, one per sentence or table row, embedded with all-MiniLM-L6-v2 (Reimers and Gurevych, 2019), and the eight chunks most similar to the question are retrieved by exact cosine search. A generator receives a fixed system prompt, two worked examples and the retrieved evidence, and must write its reasoning, a final answer and a self-reported confidence. A rule-based verifier may trigger one retry when the response's own arithmetic does not check out. Decoding is greedy unless stated. Generators are Qwen2.5-1.5B, -3B, -7B and -14B-Instruct (Qwen Team, 2024), SmolLM2-1.7B-Instruct (Allal et al., 2025) and Llama-3.1-8B-Instruct. The corrupted-evidence tests use the 3B, 7B, 14B and Llama models; the 1.5B and SmolLM2 runs enter only the error-detection and retrieval-miss analyses. The 1.5B, 3B and SmolLM2 models run in float32 on CPU; the 7B, 8B and 14B models run as 4-bit GGUF builds (Q4_K_M) through llama.cpp (Gerganov and contributors, 2023), which also resembles the 4-bit setting of earlier runs of this project.

3.2 Data

FinQA: numeric questions from the dev and test splits (the pipeline treats yes/no questions as non-numeric and excludes them from accuracy). **TAT-QA**: the arithmetic questions with a numeric answer, converted to the same item format; TAT-QA's own test labels are hidden, so its train file serves as our *dev* split and its dev file as our *test* split. Percentage answers are compared as percentages. Accuracy uses a 2% relative (or 0.01 absolute) tolerance on the final answer.

3.3 Measures under test

Numeric-overlap support (NESS) the share of the numbers in the response's reasoning and answer that occur in the supplied evidence. A response is flagged as hallucinated when this falls below 0.5. The model's own confidence statement is excluded from the count (Section 5).

Entailment support each number-bearing, non-arithmetic sentence of the reasoning is the hypothesis and the evidence, in windows of at most 280 words, is the premise. We take the maximum entailment probability over windows from the cross-encoder nli-deberta-v3-small (He et al., 2021) and report the share of claims with probability at least 0.5.

Lexical groundedness the share of the reasoning's content words (four or more letters) found in the evidence. A cheap stand-in for overlap-style faithfulness scoring.

LLM-judge faithfulness a RAGAs-style score from a language model. The judge, a different 8B model from the generator (Llama-3.1-8B, or Qwen2.5-7B for the Llama generator, 4-bit through llama.cpp), is shown the evidence and the response, lists the factual claims of the response, and marks each SUPPORTED or UNSUPPORTED by the evidence. The score is the share of claims marked SUPPORTED. The judge sees the same evidence as the generator, so it is a reference-free check of the same kind. To bound its cost it is run on the clean and fully edited evidence only.

Hallucination flag, self-reported confidence as defined above.

Two further quantities are *not* reference-free and serve as reference points: accuracy (needs the gold answer) and support *against the clean evidence* (needs the uncorrupted evidence). ONRS $= 0.30\,\text{acc} + 0.30\,\text{NESS} + 0.20\,(1 - \text{hallucination rate}) + 0.20\,(1 - \text{ECE})$, with the ECE computed on the verbalized confidence.

3.4 Evidence-side checks

The measures above look at the response. Three checks, implemented in `framework/evidence_consistency.py`, look at the evidence instead and differ in what they need. **Redundancy** needs only the evidence: the share of its numbers that occur on at least two different lines (clean filings state a figure in a table and again in the text). **Numbers vs a trusted copy** needs a trusted copy of the source: the share of the evidence's numbers that occur in it. **Cells vs a trusted table** needs a trusted *structured* copy: the share of the evidence's table cells, as (row label, column header, value) triples, that match it. We also tried arithmetic self-consistency (whether a figure is the sum, difference or percent change of two figures of one row; a figure in the text is tested against each row) and report it as a negative result.

3.5 Manipulations

Corrupted evidence. After retrieval, a fraction $p \in \{0, 25, 50, 75, 100\}\%$ of the numeric tokens

in the evidence is replaced by a wrong value (the original multiplied by 0.5, 0.8, 1.25, 1.6, 2 or 3, chosen deterministically per token), preserving currency signs, percent signs, thousands separators and decimals; four-digit years are left alone. Every other part of the pipeline is unchanged. **Realistic evidence errors.** Scaling every number is a blunt instrument. Two further manipulations keep every number that the document really contains and only misplace it, as a mis-parsed table or an off-by-one column does: *period swap* rotates the values of a table row across its period headers ("2015: 5829 | 2014: 5735" becomes "2015: 5735 | 2014: 5829"), and *row swap* exchanges the values of a table row with those of the next table row in the evidence. The level is the share of eligible table rows affected (0, 50 and 100%); level 0 is the clean evidence. **Retrieval miss.** A natural condition: an item is a *miss* when the retrieved evidence lacks at least one number of the dataset's gold evidence (for FinQA, the annotated gold facts; for TAT-QA, the operands of the annotated derivation, with scale constants and averaging divisors removed).

3.6 Statistics

Every interval is a 95% percentile bootstrap over items (2000 resamples; the rows belonging to one item stay together). The detection tasks are scored with AUROC, the probability that a randomly chosen positive case scores higher than a randomly chosen negative one (0.5 is chance).

4 Results

4.1 Corrupted evidence

Figure 1 shows how each quantity changes as more of the numbers in the retrieved evidence are replaced. Accuracy falls steeply: averaged over runs, from 0.44 to 0.22 on FinQA and from 0.53 to 0.08 on TAT-QA. Numeric-overlap support computed against the evidence the model was shown barely changes (FinQA 0.78 to 0.77, TAT-QA 0.78 to 0.78), as does entailment support and the hallucination flag. The reason is mechanical: the model copies the corrupted figures it was shown into its reasoning, and those figures are in the evidence. Support computed against the *clean* evidence falls from 0.78 to 0.56 on FinQA and from 0.78 to 0.65 on TAT-QA, because the copied figures are not in the clean evidence.

Table 1 gives the paired change per item from 0% to 100% corruption. Accuracy falls by 0.19 to 0.54 depending on run, and support against the clean evidence by 0.28 to 0.43. The gold-free measures move by at most a few points. For the larger FinQA models (Qwen2.5-7B and 14B) numeric-overlap and entailment support decline slightly but reliably (about -0.03 to -0.04 and -0.08 to -0.11), an order of magnitude less than accuracy; in every other run their changes are indistinguishable from zero or of the same small size.

The question is whether a measure can *tell* the two situations apart, not only whether its average moves. Table 2 reports the AUROC for separating a response generated from fully corrupted evidence from one generated from the clean evidence of the same item. Every gold-free measure is close to chance: pooled over runs, numeric-overlap support gives 0.54 [0.52, 0.56] on FinQA and 0.52 [0.49, 0.54] on TAT-QA, entailment support 0.55 and 0.50, and the hallucination flag 0.51 and 0.50. No single run exceeds 0.60 for any gold-free measure (the best, entailment support for Qwen2.5-14B, is 0.60 with interval 0.55 to 0.65). The model-based judge is no better: 0.50 [0.47, 0.53] pooled on FinQA and 0.50 [0.46, 0.54] on TAT-QA, and between 0.46 and 0.55 in every run. It reads the corrupted evidence, finds the response consistent with it, and has nothing else to go on. Accuracy, which needs the gold answer, reaches 0.68 and 0.73 (it is limited because many clean responses are already wrong), and support against the clean evidence reaches 0.92 and 0.88. The information that distinguishes the two cases is in the pair (response, clean evidence), which a deployed system does not have.

The composite behaves accordingly. Its correlation with the corruption level over the five levels is between -0.90 and -0.97 in every run, with intervals that exclude zero, and it comes entirely from the accuracy term and from the calibration term, which is a function of accuracy. Replacing numeric-overlap support by support against the clean evidence in the composite moves its correlation with the corruption level only from the range above to between -0.95 and -0.99, because the composite is already dominated by accuracy. So the composite does fall with corrupted evidence, but only through the terms that need the gold answer.

4.2 Realistic evidence errors

Scaling every number is a blunt error. The realistic manipulations keep every number the document contains and only misplace it (Section 3). They still hurt the model, more on TAT-QA, where a ta-

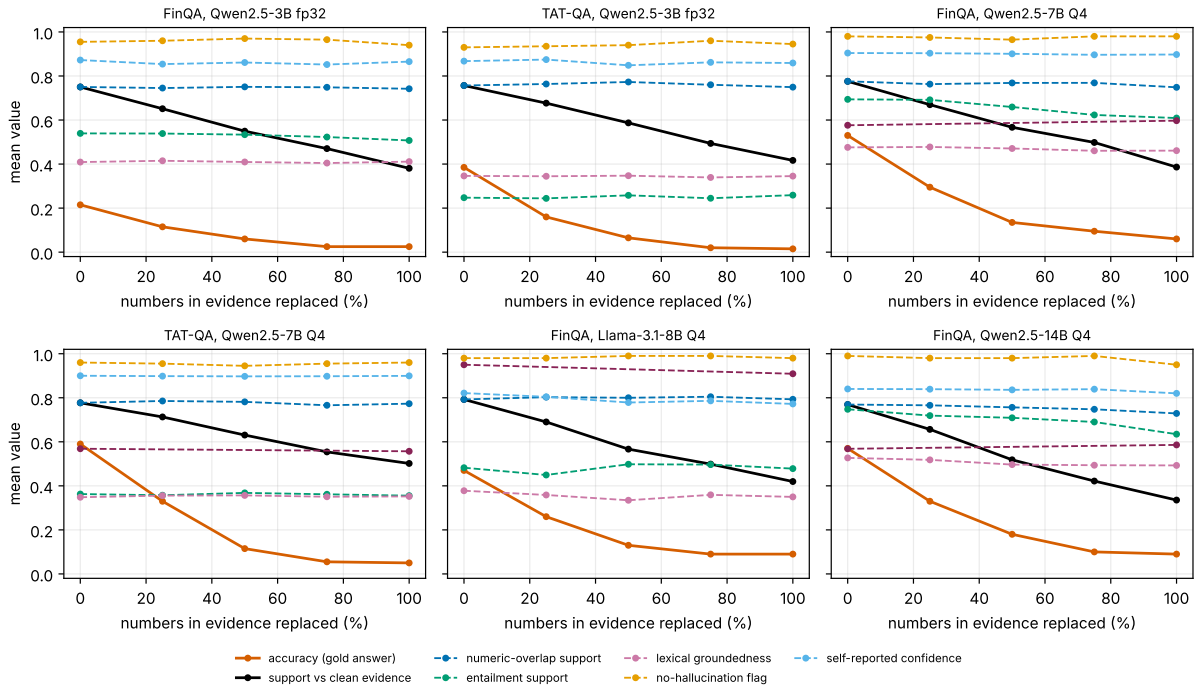


Figure 1: Mean value of each measure against the share of numbers in the retrieved evidence that were replaced by wrong ones (dev and test runs averaged within a dataset and model; 100 items per run, 200 for Qwen2.5-7B). Solid lines need the gold answer or the clean evidence; dashed lines use only the response and the evidence shown to the model.

Run	n	accuracy†	numeric-overlap support	entailment support	hallucination flag	support vs clean‡
FinQA / Qwen2.5-3B fp32 dev	100	-0.19 [-0.26, -0.12]	+0.00 [-0.02, +0.03]	+0.02 [-0.04, +0.08]	+0.01 [-0.04, +0.06]	-0.36 [-0.40, -0.32]
FinQA / Qwen2.5-3B fp32 test	100	-0.19 [-0.27, -0.11]	-0.02 [-0.07, +0.02]	-0.07 [-0.13, -0.01]	+0.02 [-0.03, +0.07]	-0.38 [-0.42, -0.33]
TAT-QA / Qwen2.5-3B fp32 dev	100	-0.37 [-0.47, -0.28]	+0.00 [-0.03, +0.04]	+0.04 [-0.02, +0.10]	-0.02 [-0.06, +0.03]	-0.35 [-0.39, -0.32]
TAT-QA / Qwen2.5-3B fp32 test	100	-0.37 [-0.46, -0.27]	-0.02 [-0.04, +0.01]	-0.02 [-0.08, +0.03]	-0.01 [-0.04, +0.02]	-0.33 [-0.36, -0.30]
FinQA / Qwen2.5-7B Q4 test	200	-0.47 [-0.54, -0.40]	-0.03 [-0.04, -0.01]	-0.08 [-0.13, -0.03]	+0.00 [-0.03, +0.03]	-0.39 [-0.41, -0.36]
TAT-QA / Qwen2.5-7B Q4 test	200	-0.54 [-0.61, -0.47]	-0.00 [-0.02, +0.01]	-0.01 [-0.05, +0.03]	+0.00 [-0.03, +0.03]	-0.28 [-0.30, -0.25]
FinQA / Llama-3.1-8B Q4 test	100	-0.38 [-0.48, -0.28]	+0.00 [-0.03, +0.03]	+0.00 [-0.07, +0.07]	+0.00 [-0.03, +0.03]	-0.37 [-0.41, -0.34]
FinQA / Qwen2.5-14B Q4 test	100	-0.48 [-0.58, -0.37]	-0.04 [-0.07, -0.01]	-0.10 [-0.16, -0.05]	+0.04 [+0.01, +0.08]	-0.43 [-0.47, -0.40]

Table 1: Change from 0% to 100% corruption, paired within item, with 95% bootstrap intervals over items. † needs the gold answer. ‡ needs the clean evidence.

ble row is nearly always in the retrieved evidence (the evidence of 86 to 93 of 100 items is changed), than on FinQA, where only 23 to 30 of 100 items have an affected table row. For Qwen2.5-7B, accuracy on TAT-QA falls from 0.62 to 0.15 when periods are swapped and to 0.08 when rows are swapped; on FinQA it falls from 0.53 to 0.46 and 0.39. Llama-3.1-8B, a second generator family, behaves alike: on TAT-QA from 0.56 to 0.16 and 0.09, on FinQA from 0.47 to 0.45 and 0.41. The evidence is the same for both generators, so the shares of items edited coincide.

Table 3 gives the AUROC for telling the fully edited evidence from the clean evidence. No response-side measure notices: across the eight runs they lie between 0.43 and 0.55 (the model-based judge, run for the Qwen generator only, between 0.49 and 0.55). The two lowest values, numeric-overlap support (0.43) and lexical groundedness (0.45) on TAT-QA rows swapped for Llama, are slight reversals, not detections. More important is the last column. Support measured against the clean evidence, which separated scaled numbers almost perfectly (AUROC 0.84 to 0.94 in Table 2), is between 0.43 and 0.52 here, because every number in the edited evidence still occurs in the clean evidence. A verifier that compares the *numbers* of the evidence with a trusted copy is blind to this class of error.

4.3 Detecting wrong answers on clean evidence

On clean evidence the gold-free measures should be useful if they carry information about errors.

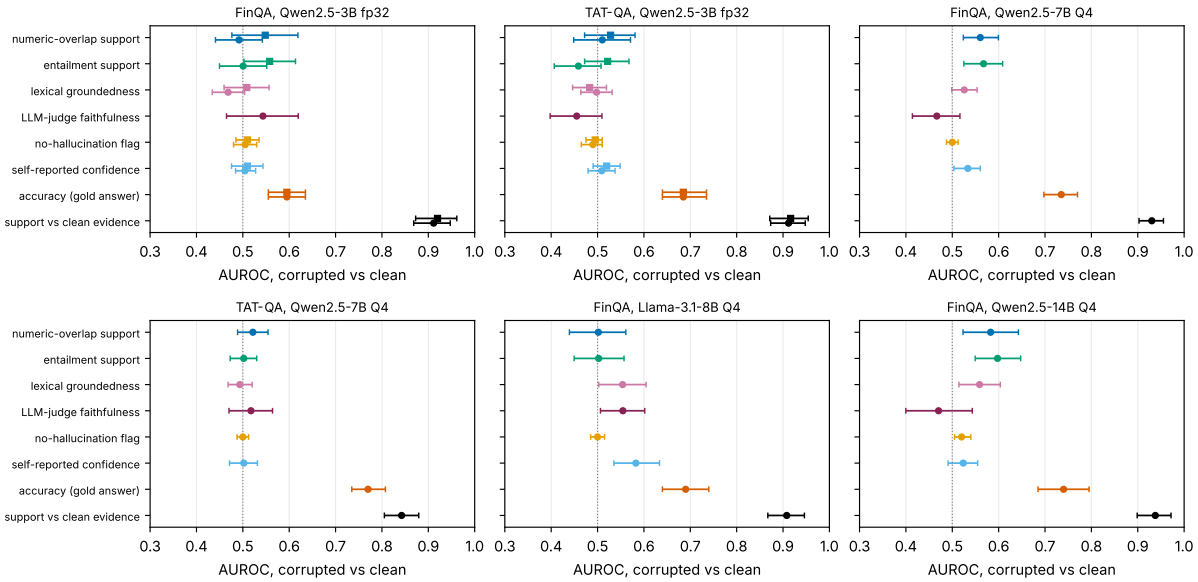


Figure 2: AUROC for telling a response produced from 100%-corrupted evidence from one produced from clean evidence (95% bootstrap intervals over items; one marker per run). The dotted line is chance.

Run	n	numeric-overlap support	entailment support	lexical groundedness	LLM-judge faithfulness	no-hallucination flag	self-reported confidence	accuracy†	support vs clean‡
FinQA / Qwen2.5-3B fp32 dev	100	0.49 [0.44, 0.54]	0.50 [0.45, 0.55]	0.47 [0.43, 0.50]	–	0.51 [0.48, 0.53]	0.50 [0.48, 0.53]	0.59 [0.56, 0.64]	0.91 [0.87, 0.95]
FinQA / Qwen2.5-3B fp32 test	100	0.55 [0.48, 0.62]	0.56 [0.50, 0.61]	0.51 [0.46, 0.56]	0.54 [0.46, 0.62]	0.51 [0.48, 0.54]	0.51 [0.48, 0.54]	0.59 [0.56, 0.64]	0.92 [0.87, 0.96]
TAT-QA / Qwen2.5-3B fp32 dev	100	0.51 [0.45, 0.57]	0.46 [0.41, 0.51]	0.50 [0.46, 0.53]	–	0.49 [0.47, 0.51]	0.51 [0.48, 0.54]	0.69 [0.64, 0.73]	0.91 [0.87, 0.95]
TAT-QA / Qwen2.5-3B fp32 test	100	0.53 [0.47, 0.58]	0.52 [0.47, 0.57]	0.48 [0.45, 0.52]	0.46 [0.40, 0.51]	0.49 [0.47, 0.51]	0.52 [0.49, 0.55]	0.69 [0.64, 0.73]	0.92 [0.87, 0.95]
FinQA / Qwen2.5-7B Q4 test	200	0.56 [0.52, 0.60]	0.57 [0.53, 0.61]	0.53 [0.50, 0.55]	0.47 [0.41, 0.52]	0.50 [0.49, 0.51]	0.53 [0.50, 0.56]	0.73 [0.70, 0.77]	0.93 [0.90, 0.96]
TAT-QA / Qwen2.5-7B Q4 test	200	0.52 [0.49, 0.55]	0.50 [0.47, 0.53]	0.49 [0.47, 0.52]	0.52 [0.47, 0.56]	0.50 [0.49, 0.51]	0.50 [0.47, 0.53]	0.77 [0.73, 0.81]	0.84 [0.81, 0.88]
FinQA / Llama-3.1-8B Q4 test	100	0.50 [0.44, 0.56]	0.50 [0.45, 0.56]	0.55 [0.50, 0.60]	0.55 [0.51, 0.60]	0.50 [0.48, 0.52]	0.58 [0.54, 0.63]	0.69 [0.64, 0.74]	0.91 [0.87, 0.95]
FinQA / Qwen2.5-14B Q4 test	100	0.58 [0.52, 0.64]	0.60 [0.55, 0.65]	0.56 [0.51, 0.60]	0.47 [0.40, 0.54]	0.52 [0.51, 0.54]	0.52 [0.49, 0.55]	0.74 [0.69, 0.80]	0.94 [0.90, 0.97]
FinQA pooled	600	0.54 [0.52, 0.56]	0.55 [0.53, 0.57]	0.52 [0.51, 0.54]	0.50 [0.47, 0.53]	0.51 [0.50, 0.51]	0.53 [0.51, 0.54]	0.68 [0.66, 0.70]	0.92 [0.91, 0.94]
TAT-QA pooled	400	0.52 [0.49, 0.54]	0.50 [0.47, 0.52]	0.49 [0.48, 0.51]	0.50 [0.46, 0.54]	0.50 [0.49, 0.51]	0.51 [0.49, 0.53]	0.73 [0.70, 0.75]	0.88 [0.85, 0.90]

Table 2: AUROC (95% interval) for separating 100%-corrupted from clean evidence. † needs the gold answer. ‡ needs the clean evidence. Pooled rows combine the runs above them.

Table 4 shows they carry little. Pooled over runs, AUROC for separating correct from incorrect answers is 0.46 for numeric-overlap support, 0.57 for entailment support, 0.55 for lexical groundedness, 0.51 for the hallucination flag and 0.55 for verbalized confidence on FinQA (TAT-QA: 0.56, 0.60, 0.55, 0.53, 0.64). Individual runs deviate from 0.5 in both directions. Numeric-overlap support is *below* 0.5 for the larger models (0.36 and 0.39 for Qwen2.5-7B on FinQA and TAT-QA, 0.37 for Qwen2.5-14B and 0.27 for Llama-3.1-8B on FinQA) and above it for the 3B model on TAT-QA (0.71 on dev, 0.74 on test): correct answers contain a computed result that is, by construction, not in the evidence, so how much weight such a score carries depends on how often the model computes versus copies. Verbalized confidence is nearly constant for the smaller models (most responses say 0.9) and uninformative there, but for the larger models it carries some signal (AUROC 0.62 to 0.73 for Qwen2.5-7B and 14B and Llama-3.1-8B), and entailment support reaches 0.70 for Llama-3.1-8B. On TAT-QA the pooled AUROCs for entailment support, lexical groundedness and confidence are 0.60, 0.55 and 0.64, modest but above chance; on FinQA all pooled values stay between 0.46 and 0.57. The model-based judge does no better than these cheaper scores: pooled 0.55 on FinQA and 0.61 on TAT-QA, and between 0.51 and 0.67 per run (it was run on the clean evidence of eight runs). The lexical baseline performs no better than the entailment or overlap scores either, so the weakness is not particular to the numeric-overlap design.

4.4 Real retrieval misses

Corruption is synthetic. A natural failure of the same kind occurs when the retriever does not surface the numbers the answer needs. Table 5 compares items by whether the retrieved evidence contains every number of the gold evidence. The miss rate is 16% on FinQA and 25% on TAT-QA. Accuracy is much lower on misses than on hits. Whether a gold-free measure notices is inconsistent. Pooled over FinQA runs, numeric-overlap support gives

Run	Error	n	numeric-overlap support	entailment support	lexical groundedness	LLM-judge faithfulness	no-hallucination flag	self-reported confidence	accuracy†	numbers vs clean‡
FinQA / Qwen2.5-7B Q4	periods swapped	100	0.50 [0.48, 0.52]	0.51 [0.50, 0.53]	0.50 [0.48, 0.51]	0.51 [0.46, 0.56]	0.50 [0.50, 0.50]	0.49 [0.47, 0.50]	0.54 [0.51, 0.56]	0.50 [0.48, 0.52]
FinQA / Qwen2.5-7B Q4	rows swapped	100	0.50 [0.48, 0.52]	0.49 [0.46, 0.51]	0.52 [0.50, 0.54]	0.54 [0.49, 0.59]	0.50 [0.50, 0.50]	0.50 [0.49, 0.51]	0.57 [0.54, 0.60]	0.50 [0.48, 0.52]
TAT-QA / Qwen2.5-7B Q4	periods swapped	100	0.50 [0.46, 0.54]	0.52 [0.49, 0.56]	0.50 [0.47, 0.54]	0.55 [0.49, 0.61]	0.50 [0.48, 0.52]	0.51 [0.48, 0.55]	0.73 [0.69, 0.79]	0.50 [0.46, 0.54]
TAT-QA / Qwen2.5-7B Q4	rows swapped	100	0.52 [0.46, 0.57]	0.53 [0.49, 0.57]	0.50 [0.45, 0.55]	0.49 [0.43, 0.56]	0.50 [0.47, 0.53]	0.53 [0.49, 0.58]	0.77 [0.72, 0.82]	0.52 [0.47, 0.58]
FinQA / Llama-3.1-8B Q4	periods swapped	100	0.50 [0.47, 0.52]	0.48 [0.46, 0.50]	0.49 [0.47, 0.51]	–	0.50 [0.50, 0.50]	0.50 [0.48, 0.52]	0.51 [0.48, 0.54]	0.50 [0.47, 0.52]
FinQA / Llama-3.1-8B Q4	rows swapped	100	0.50 [0.47, 0.52]	0.48 [0.45, 0.51]	0.50 [0.48, 0.52]	–	0.50 [0.50, 0.50]	0.49 [0.47, 0.51]	0.53 [0.50, 0.56]	0.50 [0.47, 0.52]
TAT-QA / Llama-3.1-8B Q4	periods swapped	100	0.50 [0.45, 0.54]	0.50 [0.46, 0.53]	0.49 [0.43, 0.54]	–	0.50 [0.50, 0.50]	0.52 [0.49, 0.56]	0.70 [0.65, 0.75]	0.50 [0.45, 0.54]
TAT-QA / Llama-3.1-8B Q4	rows swapped	100	0.43 [0.38, 0.49]	0.49 [0.45, 0.54]	0.45 [0.40, 0.50]	–	0.49 [0.48, 0.50]	0.54 [0.49, 0.58]	0.73 [0.69, 0.78]	0.43 [0.38, 0.49]

Table 3: AUROC (95% interval) for separating fully edited from clean evidence when table rows have their periods or values swapped (Qwen2.5-7B and Llama-3.1-8B, 100 items per row; the judge was run for the Qwen runs only). † needs the gold answer. ‡ needs the clean evidence; it compares numbers only.

Run	n	Acc.	numeric-overlap support	entailment support	lexical groundedness	LLM-judge faithfulness	no-hallucination flag	self-reported confidence
FinQA / Qwen2.5-3B fp32 dev	100	0.22	0.60 [0.47, 0.73]	0.49 [0.35, 0.62]	0.52 [0.39, 0.65]	–	0.53 [0.51, 0.56]	0.53 [0.49, 0.59]
FinQA / Qwen2.5-3B fp32 test	97	0.22	0.47 [0.33, 0.61]	0.57 [0.43, 0.71]	0.62 [0.48, 0.74]	0.57 [0.44, 0.69]	0.50 [0.44, 0.54]	0.50 [0.47, 0.53]
TAT-QA / Qwen2.5-3B fp32 dev	100	0.38	0.71 [0.60, 0.81]	0.55 [0.44, 0.67]	0.51 [0.39, 0.62]	–	0.55 [0.50, 0.60]	0.53 [0.49, 0.58]
TAT-QA / Qwen2.5-3B fp32 test	100	0.39	0.74 [0.64, 0.84]	0.52 [0.42, 0.63]	0.54 [0.42, 0.65]	0.67 [0.56, 0.78]	0.54 [0.51, 0.58]	0.53 [0.48, 0.58]
FinQA / Qwen2.5-7B Q4 test	193	0.55	0.36 [0.27, 0.43]	0.52 [0.44, 0.60]	0.51 [0.43, 0.60]	0.55 [0.47, 0.64]	0.50 [0.48, 0.52]	0.62 [0.57, 0.67]
TAT-QA / Qwen2.5-7B Q4 test	200	0.59	0.39 [0.30, 0.48]	0.63 [0.56, 0.71]	0.59 [0.51, 0.67]	0.57 [0.49, 0.65]	0.52 [0.49, 0.55]	0.73 [0.67, 0.79]
FinQA / Qwen2.5-3B fp32 dev	149	0.26	0.64 [0.54, 0.74]	0.53 [0.42, 0.63]	0.48 [0.38, 0.58]	–	0.53 [0.51, 0.56]	0.53 [0.49, 0.56]
FinQA / Qwen2.5-3B fp32 test	145	0.21	0.52 [0.39, 0.64]	0.55 [0.44, 0.67]	0.57 [0.43, 0.69]	–	0.51 [0.47, 0.54]	0.53 [0.49, 0.59]
FinQA / Qwen2.5-1.5B fp32 dev	100	0.40	0.44 [0.32, 0.56]	0.51 [0.39, 0.62]	0.46 [0.34, 0.59]	–	0.52 [0.47, 0.57]	0.53 [0.44, 0.62]
FinQA / Qwen2.5-1.5B fp32 test	97	0.34	0.48 [0.36, 0.60]	0.50 [0.39, 0.62]	0.48 [0.37, 0.61]	–	0.54 [0.51, 0.57]	0.55 [0.45, 0.65]
FinQA / SmolLM2-1.7B fp32 dev	100	0.16	0.32 [0.20, 0.46]	0.42 [0.26, 0.60]	0.36 [0.23, 0.51]	–	0.47 [0.37, 0.55]	0.50 [0.50, 0.50]
FinQA / Llama-3.1-8B Q4 test	97	0.48	0.27 [0.18, 0.38]	0.70 [0.60, 0.80]	0.55 [0.43, 0.67]	0.56 [0.50, 0.63]	0.48 [0.44, 0.50]	0.65 [0.56, 0.73]
FinQA / Qwen2.5-14B Q4 test	97	0.59	0.37 [0.25, 0.49]	0.55 [0.44, 0.65]	0.48 [0.35, 0.61]	0.51 [0.39, 0.63]	0.49 [0.47, 0.50]	0.63 [0.56, 0.69]
FinQA pooled	1175	0.35	0.46 [0.43, 0.49]	0.57 [0.53, 0.60]	0.55 [0.51, 0.58]	0.55 [0.49, 0.60]	0.51 [0.50, 0.52]	0.55 [0.54, 0.57]
TAT-QA pooled	400	0.49	0.56 [0.50, 0.62]	0.60 [0.55, 0.65]	0.55 [0.50, 0.61]	0.61 [0.55, 0.67]	0.53 [0.51, 0.56]	0.64 [0.60, 0.67]

Table 4: AUROC for separating correct from incorrect answers under clean evidence (numeric questions only; 95% bootstrap intervals over items).

AUROC 0.47, entailment 0.56 and lexical groundedness 0.49. On TAT-QA the pooled values are 0.55, 0.63 and 0.53, and verbalized confidence reaches 0.68, driven by the larger models (0.76 for Qwen2.5-7B on TAT-QA, 0.79 for Llama-3.1-8B on FinQA). The judge gives 0.58 on FinQA and 0.61 on TAT-QA pooled, and between 0.51 and 0.71 per run. Numeric-overlap support detects misses for the 3B model on TAT-QA (0.64 on dev, 0.69 on test), plausibly because with the needed figures absent the model substitutes numbers that are then not in the evidence, but the effect reverses for the larger models (0.37 to 0.44). This is a different failure from corrupted evidence in one respect: *missing* evidence can leave some trace in the response or in the model's own confidence, whereas *wrong* evidence that the model copies leaves almost none. The traces are model-dependent, and these associations are confounded with difficulty (a question whose evidence is missing is also a harder question). Accuracy itself separates hits from misses at least as well (pooled FinQA 0.68, TAT-QA 0.73).

4.5 Checking the evidence itself

If response-side measures cannot see wrong evidence, what can? Table 6 reports the checks of Section 3.4 on every edit we tried. The checks look at the evidence only, so each row is one dataset and edit and does not depend on the generator.

A check that needs only the evidence works weakly, and only for some errors. Redundancy, the share of numbers that occur on at least two lines, separates scaled evidence from clean evidence with AUROC 0.66 on FinQA and 0.68 on TAT-QA: clean filings state a figure in a table and again in the text, and independently scaled copies stop matching. It is at 0.50 for swapped periods and rows, where the numbers do not change. A second idea, that figures should be sums, differences or percent changes of figures in the same row, did not work at all (AUROC 0.44 to 0.50 on the scaled-number runs): about one number in ten is "explained" in clean and in corrupted evidence alike.

Checks against a trusted source work, if they compare the right thing. Comparing numbers with a trusted copy separates scaled evidence perfectly (AUROC 1.00) and swapped evidence not at all (0.50, 0.50). Comparing table cells, as (row label, column header, value) triples, with a trusted structured copy catches both: 0.87 and 0.97 on FinQA and 0.96 and 0.99 on TAT-QA for the swaps. The FinQA values are lower because the check is defined on only a third of the items (those whose retrieved evidence contains table cells) and because only a quarter to a third of items had an edit to find.

5 Correcting the earlier version

This study began as an evaluation of the ONRS composite on FinQA using 4-bit models on a consumer GPU. Re-running it with more items and a

Run	n	Miss	Acc. hit / miss	numeric-overlap support	entailment support	lexical groundedness	LLM-judge faithfulness	no-hallucination flag	self-reported confidence	accuracy†
TAT-QA / Qwen2.5-3B fp32 dev	100	0.27	0.48 / 0.11	0.64 [0.51, 0.78]	0.52 [0.40, 0.63]	0.50 [0.36, 0.62]	–	0.57 [0.49, 0.65]	0.57 [0.48, 0.68]	0.68 [0.60, 0.76]
TAT-QA / Qwen2.5-3B fp32 test	100	0.20	0.45 / 0.15	0.69 [0.54, 0.83]	0.63 [0.51, 0.73]	0.51 [0.37, 0.65]	0.64 [0.49, 0.78]	0.53 [0.47, 0.61]	0.58 [0.45, 0.71]	0.65 [0.55, 0.74]
FinQA / Qwen2.5-7B Q4 test	200	0.18	0.61 / 0.16	0.39 [0.29, 0.49]	0.54 [0.44, 0.65]	0.55 [0.43, 0.66]	0.56 [0.45, 0.66]	0.49 [0.47, 0.50]	0.63 [0.54, 0.72]	0.73 [0.65, 0.79]
TAT-QA / Qwen2.5-7B Q4 test	200	0.26	0.74 / 0.15	0.44 [0.33, 0.56]	0.69 [0.61, 0.76]	0.55 [0.47, 0.64]	0.60 [0.51, 0.69]	0.54 [0.50, 0.58]	0.76 [0.68, 0.83]	0.79 [0.73, 0.86]
FinQA / Qwen2.5-3B fp32 dev	150	0.13	0.29 / 0.00	0.53 [0.38, 0.68]	0.60 [0.49, 0.73]	0.48 [0.34, 0.63]	–	0.53 [0.47, 0.62]	0.61 [0.52, 0.72]	0.65 [0.61, 0.68]
FinQA / Qwen2.5-3B fp32 test	150	0.20	0.23 / 0.07	0.59 [0.47, 0.71]	0.49 [0.38, 0.60]	0.43 [0.30, 0.56]	–	0.51 [0.47, 0.56]	0.47 [0.38, 0.55]	0.58 [0.52, 0.64]
FinQA / Qwen2.5-1.5B fp32 dev	100	0.12	0.44 / 0.08	0.51 [0.33, 0.69]	0.47 [0.29, 0.65]	0.43 [0.24, 0.64]	–	0.51 [0.44, 0.61]	0.36 [0.23, 0.50]	0.68 [0.57, 0.76]
FinQA / Qwen2.5-1.5B fp32 test	100	0.17	0.39 / 0.06	0.57 [0.41, 0.73]	0.58 [0.43, 0.72]	0.48 [0.34, 0.63]	–	0.54 [0.47, 0.62]	0.44 [0.31, 0.57]	0.66 [0.58, 0.73]
FinQA / SmolLM2-1.7B fp32 dev	100	0.12	0.18 / 0.00	0.47 [0.29, 0.65]	0.56 [0.43, 0.69]	0.44 [0.27, 0.62]	–	0.45 [0.42, 0.48]	0.50 [0.50, 0.50]	0.59 [0.55, 0.63]
FinQA / Llama-3.1-8B Q4 test	100	0.17	0.55 / 0.06	0.38 [0.23, 0.55]	0.69 [0.56, 0.80]	0.50 [0.36, 0.64]	0.71 [0.58, 0.83]	0.49 [0.47, 0.50]	0.79 [0.64, 0.91]	0.75 [0.66, 0.82]
FinQA / Qwen2.5-14B Q4 test	100	0.17	0.69 / 0.00	0.37 [0.23, 0.53]	0.69 [0.54, 0.82]	0.63 [0.47, 0.76]	0.51 [0.35, 0.66]	0.49 [0.48, 0.50]	0.67 [0.53, 0.80]	0.84 [0.79, 0.89]
FinQA pooled	1000	0.16	0.42 / 0.07	0.47 [0.42, 0.52]	0.56 [0.51, 0.61]	0.49 [0.44, 0.54]	0.58 [0.50, 0.65]	0.50 [0.49, 0.52]	0.56 [0.52, 0.60]	0.68 [0.65, 0.70]
TAT-QA pooled	400	0.25	0.60 / 0.14	0.55 [0.47, 0.63]	0.63 [0.57, 0.68]	0.53 [0.47, 0.59]	0.61 [0.53, 0.69]	0.54 [0.51, 0.58]	0.68 [0.63, 0.74]	0.73 [0.68, 0.77]

Table 5: Items split by whether the retrieved evidence contains every number of the gold evidence (FinQA: annotated gold facts; TAT-QA: operands of the annotated derivation). AUROC: can the measure tell complete from incomplete retrieved evidence? † needs the gold answer.

Dataset	Error	n	edited	redundancy	arithmetic	numbers vs trusted copy	cells vs trusted table§
FinQA	numbers scaled	200	100%	0.66 [0.63, 0.70]	0.46 [0.43, 0.49]	1.00 [1.00, 1.00]	1.00 [1.00, 1.00] (34%)
FinQA	periods swapped	100	23%	0.50 [0.50, 0.50]	0.50 [0.50, 0.50]	0.50 [0.50, 0.50]	0.87 [0.80, 0.94] (31%)
FinQA	rows swapped	100	30%	0.50 [0.50, 0.50]	0.50 [0.49, 0.50]	0.50 [0.50, 0.50]	0.97 [0.92, 1.00] (31%)
TAT-QA	numbers scaled	200	98%	0.68 [0.65, 0.72]	0.55 [0.52, 0.59]	1.00 [1.00, 1.00]	1.00 [1.00, 1.00] (90%)
TAT-QA	periods swapped	100	86%	0.50 [0.50, 0.50]	0.50 [0.50, 0.50]	0.50 [0.50, 0.50]	0.96 [0.93, 0.98] (94%)
TAT-QA	rows swapped	100	93%	0.50 [0.50, 0.50]	0.50 [0.49, 0.50]	0.50 [0.50, 0.50]	0.99 [0.98, 1.00] (94%)

Table 6: AUROC (95% interval) of three evidence-side checks for telling edited from clean evidence of the same item. "edited" is the share of items whose evidence text changed. § in parentheses: the share of items on which the check is defined (it needs table cells in the retrieved evidence); the AUROC is computed on those items.

test split exposed three problems, which we report because the earlier numbers are public.

1. **A scoring flaw.** The evidence-support score counted every number in the response, including the one in the model's own "Confidence: 0.9" line, which no evidence can support. Every response carried one extra unsupported number, lowering all support scores and pushing some under the hallucination threshold on that number alone. After removing the confidence statement, the hallucination rates reported earlier (0.08 in the baseline, $0.10 \rightarrow 0.20$ under noise) are 0.00 in every condition.

2. **A manipulation that did not work.** The noise experiment added up to four sentences about unrelated figures. At 150 items per split the paired per-item change in accuracy between 0 and 40% noise was $+0.007$ (95% interval -0.05 to $+0.07$) on dev and -0.027 (-0.08 to $+0.02$) on test. The injected text reached the model but did not hurt it, so a metric could not be faulted for failing to track it. Corrupting the numbers of the evidence (Section 3) is the replacement.

3. **Unstable summary statistics.** The original conclusion that the composite fails to track noise and that no weighting of its four components rescues it came from a correlation over five noise-level means and a sweep of 1,771 weightings. On the same 3B model the correlation is $+0.09$ on dev and -0.64 on test, each with an interval of about ± 0.8, and the share of weightings passing both of the original falsification tests is 0.0% on dev and 41.2% on test. The results do not support a ranking of composites.

A statement from the earlier version survives: evidence-support measures that compare a response with the supplied evidence cannot detect corrupted evidence. It was an argument then; Section 4 makes it an experiment. The full list of corrected claims is in the repository (`ERRATUM.md`).

6 Discussion

What the results do and do not say. The tests do not show that overlap, entailment or judge scores are useless. They measure faithfulness to the supplied context, and they do that job: a response that invents a number is flagged. The finding is about what that property cannot cover. When the failure is in the evidence, any score defined relative to the evidence reports "faithful" and is wrong about reliability. A composite that adds such a score to accuracy and calibration moves with corruption only through the two terms that depend on the gold answer, and so is no better than accuracy at a task where accuracy is unavailable.

What gets past the limit. Only checks that bring in something outside the response and the evidence it was given. The cheapest such signal, redundancy inside the evidence, is weak and fails when num-

bers are moved rather than changed. A trusted copy of the numbers handles changed numbers but not moved ones, which is the more realistic failure of a mis-parsed table. The check that handled every error we tried compares (row, column, value) cells with a trusted structured source, such as the XBRL record behind a filing. It needs that source to exist and the evidence to carry table structure, which limits it to part of the items (a third on FinQA). Where such a source exists the practical conclusion is simple: verify the evidence against it before generation, and do not read a faithfulness score as a reliability estimate.

Implications for practice and for evaluation. (i) A reliability pipeline needs a signal about the evidence, not only about the response. (ii) A verifier that compares numbers is not enough; compare structure. (iii) Validate an evidence-quality manipulation by its effect on the model before using it to evaluate a metric. The distractor sentences we first used would have been reported as a metric failure.

7 Conclusion

Reference-free reliability scores for numerical RAG cannot see wrong evidence, and a controlled test on two datasets and four generators from two model families confirms it: accuracy collapses while every measure that sees only the response and the supplied evidence stays near chance (AUROC at most 0.60). Realistic errors that merely misplace genuine numbers cost as much accuracy and escape even a check of the numbers against the clean evidence; only a cell-level check against a trusted structured copy catches them. The same response-side measures are weak and model-dependent at flagging wrong answers on clean evidence and real retrieval misses. Practitioners who need reliability estimates should treat such scores as faithfulness checks and add a verification of the evidence against a trusted structured source.

Limitations

The generators are open models of 1.5B to 14B parameters, run without fine-tuning; larger or finance-tuned models may behave differently, and the 14B model was run on 100 items. Accuracy is low for the smaller models (about 0.2 on FinQA for the 3B model; 0.47 to 0.62 for the 7B and larger models), which limits the power of the correct-versus-incorrect comparisons; we report intervals instead of claiming equality. The scaled-number corruption is synthetic and numeric. The realistic errors (swapped periods and rows) are closer to real parsing failures but are still injected, were run for two generators (Qwen2.5-7B and Llama-3.1-8B, 4-bit) and only on table rows, and on FinQA affect only the quarter to third of items whose retrieved evidence has an eligible table row. The cell-level check is, by construction, an upper bound: it assumes a trusted structured copy and that the evidence carries table structure. The retrieval-miss analysis defines a miss by the absence of gold numbers in the retrieved text, which can disagree with whether the answer was recoverable; the TAT-QA gold evidence is the operands of the annotated derivation because the dataset has no cell-level evidence labels. The entailment measure uses one small NLI model; the lexical baseline is a heuristic; the faithfulness judge is a single 4-bit 8B model with one prompt, run only on the clean and fully edited evidence, and a stronger judge or a different prompt may do better. It is a RAGAs-*style* score, not the RAGAs implementation. The 7B, 8B and 14B models are 4-bit builds run through a different inference engine than the float32 models, so differences between sizes are confounded with precision. We do not report a GPU 4-bit run of the smaller models. Item counts (100 to 200 per run) are modest; pooling over runs narrows the intervals but mixes models.

Reproducibility

Code, prompts, per-response outputs for every run, and the GitHub Actions workflow that regenerates them on free CPU runners are at `https://github.com/aaronsunny007/Financial-NRF`. Tables and figures in this paper are produced from the result files by `experiments/paper_analysis.py` and `experiments/paper_figures.py`.

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References

Loubna Ben Allal et al. 2025. SmolLM2: When smol goes big – data-centric training of a small language model. *arXiv preprint arXiv:2502.02737*.

Zhiyu Chen, Wenhu Chen, Charese Smiley, Sameena Shah, Iana Borova, Dylan Langdon, Reema Moussa, Matt Beane, Ting-Hao Huang, Bryan Routledge, and William Yang Wang. 2021. FinQA: A dataset of numerical reasoning over financial data. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*.

Shahul Es, Jithin James, Luis Espinosa-Anke, and Steven Schockaert. 2024. RAGAs: Automated evaluation of retrieval augmented generation. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*.

Georgi Gerganov and contributors. 2023. llama.cpp: Inference of LLaMA models in pure C/C++. https://github.com/ggml-org/llama.cpp.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. 2017. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*.

Pengcheng He, Jianfeng Gao, and Weizhu Chen. 2021. DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing. *arXiv preprint arXiv:2111.09543*. Published at ICLR 2023.

Saurav Kadavath et al. 2022. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*.

Philippe Laban, Tobias Schnabel, Paul N. Bennett, and Marti A. Hearst. 2022. SummaC: Re-visiting NLI-based models for inconsistency detection in summarization. *Transactions of the Association for Computational Linguistics*, 10.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*.

Potsawee Manakul, Adian Liusie, and Mark Gales. 2023. SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*.

Joshua Maynez, Shashi Narayan, Bernd Bohnet, and Ryan McDonald. 2020. On faithfulness and factuality in abstractive summarization. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*.

Sewon Min, Kalpesh Krishna, Xinxi Lyu, Mike Lewis, Wen-tau Yih, Pang Wei Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. 2023. FActScore: Fine-grained atomic evaluation of factual precision in long form text generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*.

Qwen Team. 2024. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*.

Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*.

Jon Saad-Falcon, Omar Khattab, Christopher Potts, and Matei Zaharia. 2024. ARES: An automated evaluation framework for retrieval-augmented generation systems. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics*.

Miao Xiong, Zhiyuan Hu, Xinyang Lu, Yifei Li, Jie Fu, Junxian He, and Bryan Hooi. 2024. Can LLMs express their uncertainty? an empirical evaluation of confidence elicitation in LLMs. In *International Conference on Learning Representations*.

Yuheng Zha, Yichi Yang, Ruichen Li, and Zhiting Hu. 2023. AlignScore: Evaluating factual consistency with a unified alignment function. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*.

Fengbin Zhu, Wenqiang Lei, Youcheng Huang, Chao Wang, Shuo Zhang, Jiancheng Lv, Fuli Feng, and Tat-Seng Chua. 2021. TAT-QA: A question answering benchmark on a hybrid of tabular and textual content in finance. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics*.